Max. Time : 129 Minutes			Total Marks-166
Q. 01 to 06	Comprehension	('–1' negative marking)	[3 Marks,3 Min.][18,18]
Q. 07 to 26	Multiple choice objective	('–1' negative marking)	[4 Marks,3 Min.][80,60]
Q. 27 to 28	Match the column	(no negative marking)	[4 Marks,3 min.][8,6]
Q. 29 to 43	Numerical Type	('–1' negative marking)	[4 Marks,3 Min.][60,45]

DPP Syllabus : Trigonometry, Sequence and Series, Statistics

Comprehension # 1 (Q.No. 1 to 2)

We know that $1 + 2 + 3 + ... = \dfrac{n(n+1)}{2} = f(n)$,

$1^2 + 2^2 + 3^2 + ... + n^2 = \dfrac{n(n+1)\ \ (2n+1)}{6} = g(n)$,

$1^3 + 2^3 + 3^3 + ... + n^3 = \left(\dfrac{n(n+1)}{2}\right)^2 = h(n)$

1. Even natural number which divides g(n) – f(n), for every $n \geq 2$, is

(A) 2 (B) 4 (C) 6 (D) none of these

2. f(n) + 3 g(n) + h(n) is divisible by 1 + 2 + 3 + ... + n

(A) only if n = 1 (B) only if n is odd (C) only if n is even (D) for all $n \in N$

Comprehension # 2 (Q.No. 3 to 4)

Let $\Delta^1 T_n = T_{n+1} - T_n$, $\Delta^2\ T_n = \Delta^1 T_{n+1} - \Delta^1 T_n$, $\Delta^3 T_n = \Delta^2 T_{n+1} - \Delta^2 T_n$, ... , and so on, where $T_1, T_2, T_3, ...T_{n-1}, T_n, T_{n+1}, ...$ are the terms of infinite G.P. whose first term is a natural number and common ratio is equal to 'r'.

3. If $\Delta^2 T_1 = 36$, then sum of all possible integral values of r is equal to :

(A) 8 (B) 4 (C) 5 (D) –2

4. Let $\sum_{n=1}^{\infty} T_n = \dfrac{7}{3}$ and $r = \dfrac{p}{7}$ then sum of squares of all possible value of p is equal to :

(A) 42 (B) 46 (C) 45 (D) 30

Comprehension # 3 (Q.No. 5 to 6)

Let a, b, c, d $\in$ R. Then the cubic equation of the type $ax^3 + bx^2 + cx + d = 0$ has either one root real or all three roots are real. But in case of trigonometric equations of the type $a \sin^3 x + b \sin^2 x + c \sin x + d = 0$ can possess several solutions depending upon the domain of x.

To solve an equation of the type $a \cos\theta + b \sin\theta = c$. The equation can be written as $\cos(\theta - \alpha) = c/\sqrt{(a^2 + b^2)}$. The solution is $\theta = 2n\pi + \alpha \pm \beta$, where $\tan \alpha = b/a$, $\cos \beta = c/\sqrt{(a^2 + b^2)}$.

5. On the domain $[-\pi, \pi]$ the equation $4\sin^3 x + 2 \sin^2 x - 2\sin x - 1 = 0$ possess

(A) only one real root (B) three real roots (C) four real roots (D) six real roots

6. In the interval $[-\pi/4, \pi/2]$, the equation, $\cos 4x + \dfrac{10 \tan x}{1 + \tan^2 x} = 3$ has

(A) no solution (B) one solution (C) two solutions (D) three solutions

7. Let $E = \cos^2 \dfrac{\pi}{7} + \cos^2 \dfrac{2\pi}{7} + \cos^2 \dfrac{3\pi}{7}$, then which of the following is/are false

(A) $\dfrac{1}{2} < E < \dfrac{3}{4}$ (B) $\dfrac{3}{4} < E < 1$ (C) $1 < E < \dfrac{3}{2}$ (D) $2 < E < 3$

8. If $S_n(\theta) = \sin 2\theta \sin^2 \theta + \dfrac{1}{2} \sin 4\theta \sin^2 2\theta + \dfrac{1}{4} \sin 8\theta \sin^2 4\theta + \ldots\ldots\ldots$ up to n terms , then

(A) $\lim\limits_{n\to\infty} S_n\left(\dfrac{\pi}{6}\right) = \dfrac{\sqrt{3}}{4}$ (B) $\lim\limits_{n\to\infty} S_n\left(\dfrac{\pi}{3}\right) = \dfrac{\sqrt{3}}{4}$ (C) $\lim\limits_{n\to\infty} S_n(\theta) = \dfrac{\sin 2\theta}{2}$ (D) $\lim\limits_{n\to\infty} S_n(\theta) = \dfrac{\sin 2\theta}{4}$

9. Which of following functions have the maximum value unity?

(A) $\sin^2 x - \cos^2 x$

(B) $\sqrt{\dfrac{6}{5}}\left(\dfrac{1}{\sqrt{2}} \sin x + \dfrac{1}{\sqrt{3}} \cos x\right)$

(C) $\cos^6 + \sin^6 x$

(D) $\cos^2 x + \sin^4 x$

10. The solution of the equation $3^{\sin 2x + 2\cos^2 x} + 3^{1 - \sin 2x + 2\sin^2 x} = 28$

(A) $(2n+1)\dfrac{\pi}{2}, n \in I$ (B) $n\pi - \dfrac{\pi}{4}, n \in I$ (C) $2n\pi$ (D) none of these

11. If $\dfrac{\sin^4 x}{2} + \dfrac{\cos^4 x}{3} = \dfrac{1}{5}$, then

(A) $\tan^2 x = \dfrac{2}{3}$

(B) $\dfrac{\sin^8 x}{8} + \dfrac{\cos^8 x}{27} = \dfrac{1}{125}$

(C) $\tan^2 x = \dfrac{1}{3}$

(D) $\dfrac{\sin^8 x}{8} + \dfrac{\cos^8 x}{27} = \dfrac{1}{125}$

12. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is(are)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

13. If a,b,c,d are four unequal positive numbers which are in A.P. then

(A) $\frac{1}{a}+\frac{1}{d}=\frac{1}{b}+\frac{1}{c}$ (B) $\frac{1}{a}+\frac{1}{d}<\frac{1}{b}+\frac{1}{c}$ (C) $\frac{1}{a}+\frac{1}{d}>\frac{1}{b}+\frac{1}{c}$ (D) $\frac{1}{b}+\frac{1}{c}>\frac{4}{a+d}$

14. If $a_1, a_2, \ldots\ldots., a_n$ are distinct terms of an A.P., then

(A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$

(C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$

15. If b_1, b_2, b_3 $(b_i > 0)$ are three successive terms of a G.P. with common ratio r, the value of r for which the inequality $b_3 > 4b_2 - 3b_1$ holds is given by

(A) $r > 3$ (B) $0 < r < 1$ (C) $r = 3.5$ (D) $r = 5.2$

16. Let $a_1, a_2, a_3, \ldots\ldots a_n$ is the sequence whose sum of first 'n' terms is represented by

$S_n = an^3 + bn^2 + cn$, $n \in N$. If $a = \frac{a_1 + a_3 - xa_2}{y}$ then

(A) H.C.F of (x,y) is 2 (B) H.C.F. of (x,y) is 3

(C) L.C.M of (x,y) is 6 (D) x + y = 8

17. In a G.P. the ratio of the sum of the first eleven terms to the sum of the last eleven terms is $\frac{1}{8}$ and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Then the number of terms of the G.P is less than.

(A) 15 (B) 43 (C) 38 (D) 56

18. For the series $S = 1 + \frac{1}{(1+3)}(1+2)^2 + \frac{1}{(1+3+5)}(1+2+3)^2 + \frac{1}{(1+3+5+7)}(1+2+3+4)^2 + \ldots..$

(A) 7^{th} term is 16 (B) 7^{th} term is 18

(C) sum of first ten terms is $\frac{505}{4}$ (D) sum of first ten term is $\frac{405}{4}$

19. If $\sum_{r=1}^{n} r(r+1) = \frac{(n+a)(n+b)(n+c)}{3}$, where $a < b < c$, then

(A) $2b = c$ (B) $a^3 - 8b^3 + c^3 = 8abc$

(C) c is prime number (D) $(a + b)^2 = 0$

20. Let $a_n = \underset{\text{n times}}{(111.....1)}$, then

(A) a_{912} is not prime (B) a_{951} is not prime (C) a_{480} is not prime (D) a_{91} is not prime

21. If $x_1, x_2,.....x_n$ are n observations such that $\sum_{i=1}^{n} x_i^2 = 400$ and $\sum_{i=1}^{n} x_i = 100$, then possible value of n cannot be

(A) 22 (B) 8 (C) 11 (D) 12

22. Let the mean and the variance of 20 observations $x_1, x_2, \ldots x_{20}$ be 15 and 9, respectively. For $\alpha \in R$, if the mean of $(x_1 + \alpha)^2, (x_2 + \alpha)^2, \ldots, (x_{20} + \alpha)^2$ is 178, then

(A) Square of the maximum value of α is 4 (B) Sum of distinct value of α is –30

(C) maximum value of α lies in [–4, 0] (D) α is –2, –20

23. For the frequency distribution :

Variate (x) :	x_1	x_2	x_3 .....x_{15}
Frequency (f) :	f_1	f_2	f_3 ....f_{15}

where $0 < x_1 < x_2 < x_3 < .... < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, the standard deviation can be :

(A) 2 (B) 1 (C) 4 (D) 6

24. The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5, 7, 10, 12, 14, 15, if other two observations is a, b, then

(A) $|a - b| = 7$ (B) $a^2 + b^2 = 169$ (C) $a = 8, b = 1$ (D) a and b are prime number

25. Consider the following frequency distribution :

Class :	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30
Frequency :	a	b	12	9	5

If mean $= \frac{309}{22}$ and median = 14, then

(A) $(a - b)^2 = 4$

(B) quadratic equation whose roots are a, b is $x^2 - 18x + 80 = 0$

(C) Common divisor of a & b is 2

(D) a + b is divisible by 9

26. Let the mean and variance of the frequency distribution

x :	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
f :	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then

(A) $\alpha + \beta = 7$ (B) mean of new data is $\frac{17}{3}$

(C) $\alpha^2 + \beta^2 = 29$ (D) mean of new data is $\frac{16}{3}$

27. **Column – I** **Column – II**

Column – I	Column – II
(A) Number of solutions of $\sin^2\theta + 3\cos\theta = 3$ in $[-\pi, \pi]$	(p) 0
(B) Number of solutions of $\sin x \cdot \tan 4x = \cos x$ in $(0, \pi)$	(q) 1
(C) Number of equation $(1 - \tan\theta)(1 + \tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$ where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	(r) 4
(D) If $[\sin x] + [\sqrt{2}\cos x] = -3$, where $x \in [0, 2\pi]$ then $[\sin 2x]$ equals (Here [.] denotes G.I.F.)	(s) 5

28. **Column – I** **Column – II**

Column – I	Column – II
(A) The value of xyz is 15/2 or 18/5 according as the series a, x, y, z, b are in an A.P. or H.P. then 'a + b' equals where a, b are positive integers.	(p) 2
(B) The value of . $2^{\frac{1}{4}}\, 4^{\frac{1}{8}}\, 8^{\frac{1}{16}} \cdot\text{- - -}\infty$ is equal to	(q) 1
(C) If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$, where $k \in N$, then k is equal to	(r) 3
(D) There are m A.M. between 1 and 31. If the ratio of the 7^{th} and $(m-1)^{th}$ means is 5 : 9, then $\frac{m}{7}$ is equal to	(s) 4

29. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm1, \pm 2$ and $\tan\theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$ is

30. If $A + B + C = \pi$, then find value of $\tan B \tan C + \tan C \tan A + \tan A \tan B - \sec A \sec B \sec C$.

31. $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = \frac{1}{e}$, where $\alpha, \beta \in [-\pi, \pi]$. Then number of ordered pairs (α, β) which satisfy both the equations.

32. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then, the value of $\frac{b}{a}$ is ______.

33. Find the number of triplets (x, y, z), where $x, y, z \in [0, 2\pi]$ satisfying the inequality $(4 + \sin 4x)(2 + \cot^2 y)(1 + \sin^4 z) \leq 12 \sin^2 z$.

34. Let $f(x) = a_0 + a_1x + a_2x^2 + ... + a_nx^n + ...$ and $\frac{f(x)}{1-x} = b_0 + b_1x + b_2x^2 + ... + b_nx^n + ...$ given that $a_0, a_1, a_2, ...$ are in G.P. If $a_0 = 1$ and $b_1 = 3$ then find value of b_{10}

35. If $\sum_{r=1}^{n} r^3 - \sum_{p=1}^{n}\sum_{m=1}^{p}\sum_{r=1}^{m} 1 = 80$, then find possible value of n.

36. For the sequence 1,22,333,4444,55555,..... Characteristic $\log_{10}(T_{100})$ is
(where T_r denotes r^{th} term of sequence)

37. If $a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + ...(-1)^{n-1}\left(\frac{3}{4}\right)^n$ and $b_n = 1 - a_n$, then find the minimum natural number n_0 such that $b_n > a_n \ \forall \ n > n_0$

38. If a, b, c are positive real numbers such that $a + b + c = 1$ then find the minimum value of $\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}$.

39. $x_1, x_2........x_{34}$ are numbers such that $x_i = x_{i+1} = 150 \ \forall \ i \in \{1,2,3,......9\}$ and
$x_{i+1} - x_i + 2 = 0 \ \forall \ i \in \{10,11,12,......33\}$, then median of $x_1, x_2,......x_{34}$ is-

40. If $\sum_{i=1}^{5}\left(x_i-10\right)=5$ and $\sum_{i=1}^{5}\left(x_i-10\right)^2=25$, then standard deviation of observations $2x_1+7$, $2x_2+7$, $2x_3+7$, $2x_4+7$ and $2x_5+7$ is equal to-

41. Coefficient of variance of a distribution is 60% and the standard deviation is 25. The arithmetic mean of the distribution is $\overline{x}$, then $3\overline{x}$ is equal to

42. The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to___________.

43. Consider the data on x taking the values 0, 2, 4, 8, ..., 2^n with frequencies ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ..., ${}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to ________.

ANSWER KAY

1.	(A)	**2.**	(D)	**3.**	(A)	**4.**	(B)	**5.**	(D)	**6.**	(C)	**7.**	(ABD)
8.	(ABC)	**9.**	(ABCD)			**10.**	(AB)	**11.**	(AB)	**12.**	(CD)	**13.**	(CD)
14.	(BD)	**15.**	(ABCD)			**16.**	(ACD)	**17.**	(BD)	**18.**	(AC)	**19.**	(ABC)
20.	(ABCD)			**21.**	(ABCD)			**22.**	(ABC)	**23.**	(ABC)	**24.**	(AB)
25.	(ABCD)			**26.**	(ABC)	**27.**	(A) $\rightarrow$ (q), (B) $\rightarrow$ (s), (C) $\rightarrow$ (r), (D) $\rightarrow$ (p)						
28.	(A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)							**29.**	(3)	**30.**	(1)	**31.**	(4)
32.	(0.5)	**33.**	(16)	**34.**	(2047)	**35.**	(4)	**36.**	(299)	**37.**	(5)	**38.**	(27)
39.	(135)	**40.**	(4)	**41.**	(125)	**42.**	(238)	**43.**	(6.00)				

Reliable INSTITUTE
Address : 'Reliable Institute', A-10 Road No.1, IPIA, Kota-324005 (Rajasthan), INDIA
visit us at: www.reliablekota.com, Email: info@reliablekota.com
Call us: +91-744-2665544